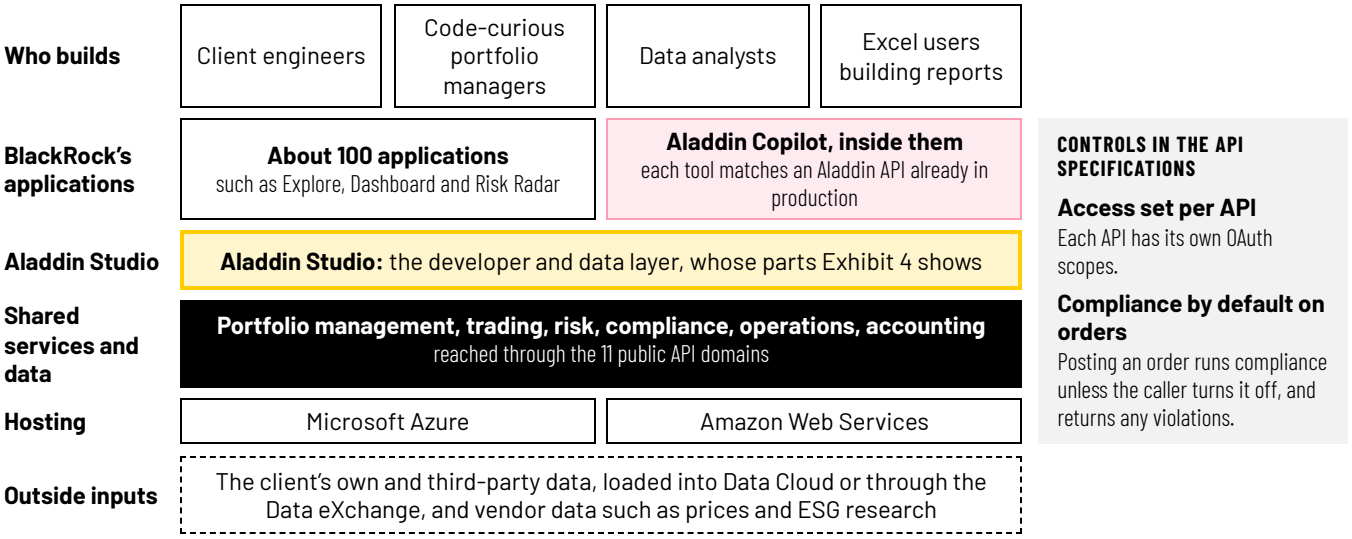
Exhibit 3. Where Studio sits in Aladdin



For illustrative purposes only. Every box is named in the sources below. The 11 domains are the packages of BlackRock's public software development kit (SDK). The specifications also name Dashboard and Risk Radar. Controls as in Exhibit 14. ■ Source: Rosales at 2:35 and 4:31, for the apps and Copilot[9]; a training recording, for Explore[27]; Woodhead at 4:49, for Studio[1]; the public specifications and SDK[6, 29]; BlackRock's Studio, APIs, ATX, ADX, Data Cloud and Aladdin Risk pages[14, 20, 30, 31, 32, 33]; the Data Cloud launch release[34]; the FY2025 10-K, for hosting[12]; Microsoft, for Azure[35]; data vendors, ICE and Sustainalytics[36, 37].

Exhibit 4. What Studio is made of

WHAT STUDIO OFFERS

APIs
Read and write Aladdin data. Built on Aladdin Graph, BlackRock's firm-wide runtime for APIs and events. Every public specification's ID begins `agraph.<domain>`, for Aladdin Graph

Event-driven workflows
Aladdin tells a program when something happens. Events run on the same Aladdin Graph runtime. No event specification is public

Aladdin SDK
Open-source Python library with typed clients for the APIs and Data Cloud, with login, retries and rate limits built in. A BlackRock plugin builder generates its 11 plugin packages from the specifications

Aladdin Data Cloud
A managed data store on Snowflake, with Fivetran moving the data

Aladdin Compute
Hosted, ready-to-use Python environments

Aladdin Tools for Excel (ATX)
An Excel plugin that pulls Aladdin data into custom reports and analyses

Aladdin Data eXchange (ADX)
Bulk file transfer to and from Aladdin, with standard extracts and loaders

Developer Portal and Data Dictionary
Catalog, guides, field dictionary, notebooks and samples, behind a client login

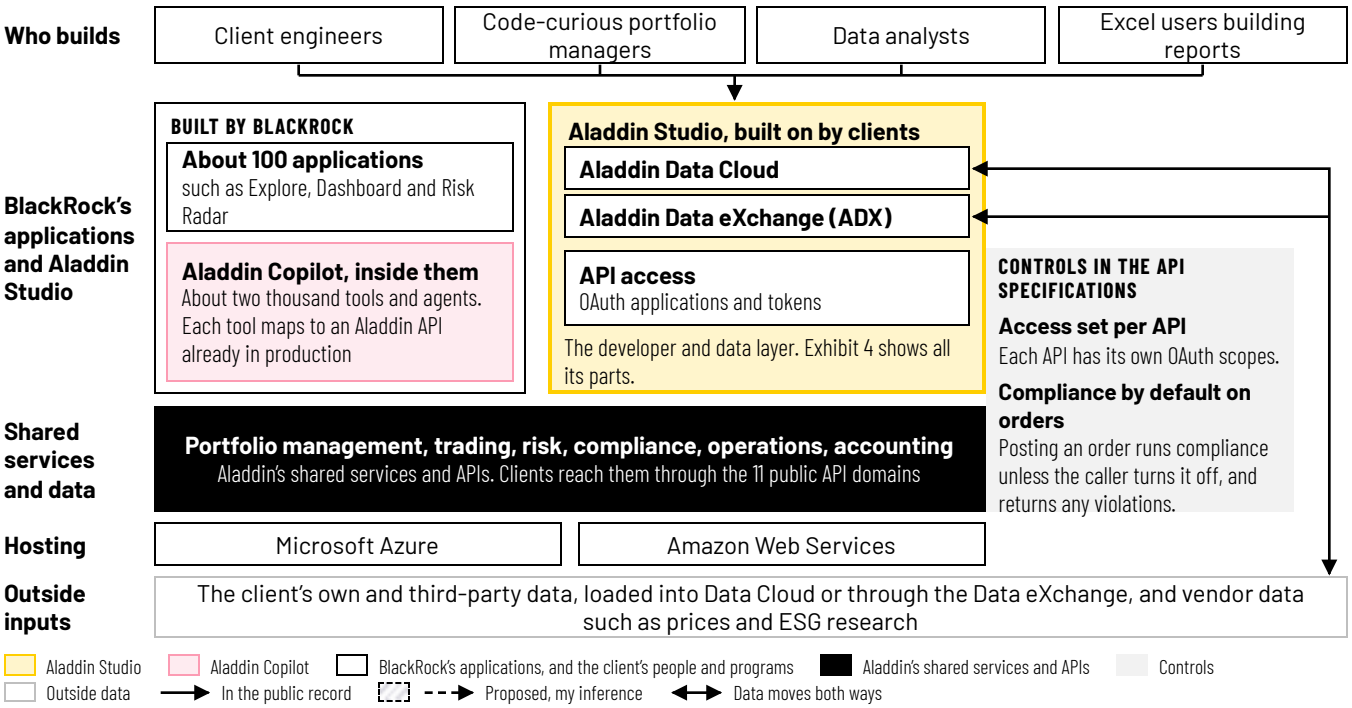
Governance
"Data, tools, and governance support", including "access management and usage monitoring"

INSIDE A CLIENT'S STUDIO WORKSPACE

- **Studio UI** The screens where tokens, applications and storage are set up
- **API tokens and OAuth applications** Sign-in for the APIs
- **S3 storage integration** File storage for SDK code, set up in a Project
- **ADC Admin Center** The screen for managing Data Cloud datasets
- **Projects** Personal or shared, each holding OAuth applications and storage
- **Studio Compute sessions** Where Aladdin Compute runs SDK code
- **Studio notifications** Alerts sent through Studio's notification APIs

■ Source: BlackRock's Studio page[14]; its APIs, ATX and ADX pages, which list Data Cloud, ADX, Aladdin Compute and ATX as "Additional Aladdin Studio resources"[30, 32, 33]; the 54 public specifications[6], for the ids, the portal link and the Admin Center; the SDK README[29], for the SDK and the workspace; Naik and Kalish[38] and the Aladdin Graph job posting[39], for Aladdin Graph; the plugin builder[40]; Woodhead[1], Goel[41], the Aladdin Risk page[31] and a BlackRock job posting[42], for events and the portal; Fivetran[28]; Snowflake's partner page, for Compute[43]; the developer portal[44]; Studio for Aladdin Risk, for access management[45]. I read Studio Compute sessions as the SDK's name for Aladdin Compute.

Exhibit 3. Where Studio sits in Aladdin



For illustrative purposes only. Every box is named in the sources below. The 11 domains are the packages of BlackRock's public software development kit (SDK). The specifications also name Dashboard and Risk Radar. Exhibit 4 opens the Studio box. Controls as in Exhibit 14. ■ Source: Rosales, for the apps and Copilot[9]; a training recording, for Explore[27]; Woodhead, for Studio[1]; the public specifications and SDK[6, 29]; BlackRock's Studio, APIs, ATX, ADX, Data Cloud and Aladdin Risk pages[14, 20, 30, 31, 32, 33]; the Data Cloud launch release[34]; the FY2025 10-K, for hosting[12]; Microsoft, for Azure[35]; data vendors, ICE and Sustainalytics[36, 37].

Exhibit 4. What Studio is made of

ALADDIN STUDIO, BUILT ON BY CLIENTS · THE YELLOW BOX IN EXHIBIT 3, OPENED UP

API access OAuth applications and tokens that let a program call Aladdin's APIs. The APIs themselves are the shared layer below	**Event-driven workflows** Aladdin tells a program when something happens. No public specification covers business events	**Aladdin SDK** Open-source Python library with typed clients for the APIs and a Snowflake connection for Data Cloud
Aladdin Data Cloud A managed data store on Snowflake	**Aladdin Compute** Hosted, ready-to-use Python environments	**Aladdin Tools for Excel (ATX)** An Excel plugin that pulls Aladdin data into custom reports and analyses
Aladdin Data eXchange (ADX) Bulk file transfer to and from Aladdin	**Developer Portal and Data Dictionary** Guides, behind a client login	**Governance** "Access management and usage monitoring"

INSIDE A CLIENT'S STUDIO WORKSPACE

- **Studio UI** The screens where tokens, applications and storage are set up
- **Projects** Personal or shared, each holding OAuth applications and storage
- **Spaces** Named alongside Projects. Not described publicly
- **ADC Admin Center** The screen for managing Data Cloud datasets
- **API tokens and OAuth applications** Sign-in for the APIs
- **Studio Compute sessions** Where Aladdin Compute runs SDK code
- **S3 storage integration** File storage for SDK code, set up in a Project
- **Studio notifications** Alerts sent through Studio's notification APIs

Shared layer: Aladdin's APIs. They read and write Aladdin data, built on Aladdin Graph, BlackRock's firm-wide runtime for APIs and events.

Colours as in Exhibit 3. The SDK tile and the workspace items are as named in the public SDK, which may not match the current product. The public SDK uses both names, Studio Compute and Aladdin Compute, for the same hosted environment. ■ Source: BlackRock's Studio page[14]; its APIs, ATX and ADX pages, which list Data Cloud, ADX, Aladdin Compute and ATX as "Additional Aladdin Studio resources"[30, 32, 33]; the 54 public specifications[6], for OAuth, the portal link and the Admin Center; the public SDK's README and source[29], for the SDK, the workspace, Spaces and both Compute names; Naik and Kalish[38] and the Aladdin Graph job posting[39], for Aladdin Graph; Woodhead[1], Goel[41], the Aladdin Risk page[31] and a BlackRock job posting[42], for events and the portal; Snowflake's partner page, for Compute[43]; the developer portal[44]; Studio for Aladdin Risk, for access management[45].

Independent analysis from public sources. Not produced, reviewed or endorsed by BlackRock.

Part 3

Directions to explore

BlackRock could present Studio as the place where a client's own AI tools act on investment data. Those tools would run under the same controls as BlackRock's own business. BlackRock has said parts of this. It sells Aladdin as the system it runs internally[9, 83]. Its speakers have said that the system of record underpins anything built with AI[10], and that the data model is being structured for agentic use[72]. I have not seen BlackRock describe Studio this way.

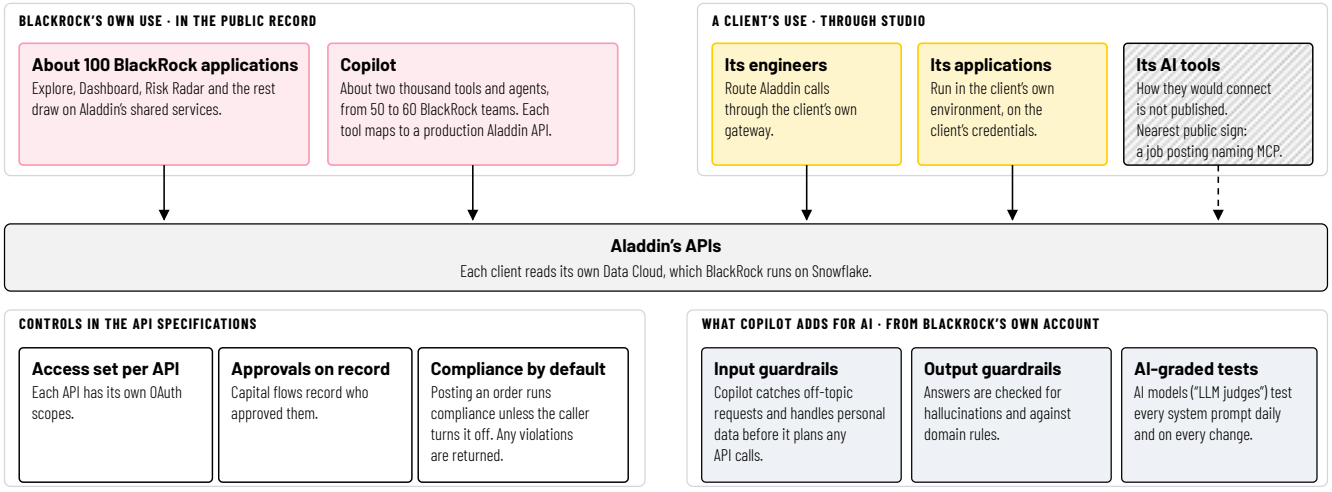
Letting AI tools act is unlikely to stay unique. SimCorp and Clearwater have announced AI agents, Arcesium an AI platform, and others will likely follow[7, 8, 76]. What BlackRock can show is its own use. About 100 of its applications carry Copilot. Each Copilot tool maps to a production Aladdin API. The team that owns each tool sets which user groups can use it[9]. Posting an order runs compliance by default. The caller can turn it off[6]. Whether turning it off needs a separate permission is not public. Copilot's guardrails are tested daily[9]. A client's AI tools would work under controls BlackRock already relies on for its own business. How those tools would connect is not public.

Buyers say AI is changing what they build. "AI and data demands" led BlackRock's webinar poll at 42 percent as the reason firms are rethinking what to build and what to buy, and buyers now ask about APIs and connectors in RFPs[10]. The poll did not ask whether clients want AI tools that act on Aladdin, and it gives no sample size.

The client base is large. BlackRock serves hundreds of Aladdin clients in 70 countries[9], and its technology services revenue was $2.0 billion in 2025[12]. A 2026 job posting for Aladdin Graph, the team behind the API layer, says it builds for BlackRock and over 200 institutions that rely on its technology[39]. BlackRock sizes Aladdin's addressable market at more than $20 billion[11]. It projects API calls to grow 17x by 2030[11]. My inference is that clients' AI tools, acting through Studio, could become a large source of those calls. How many of those institutions run their own AI tools is not public.

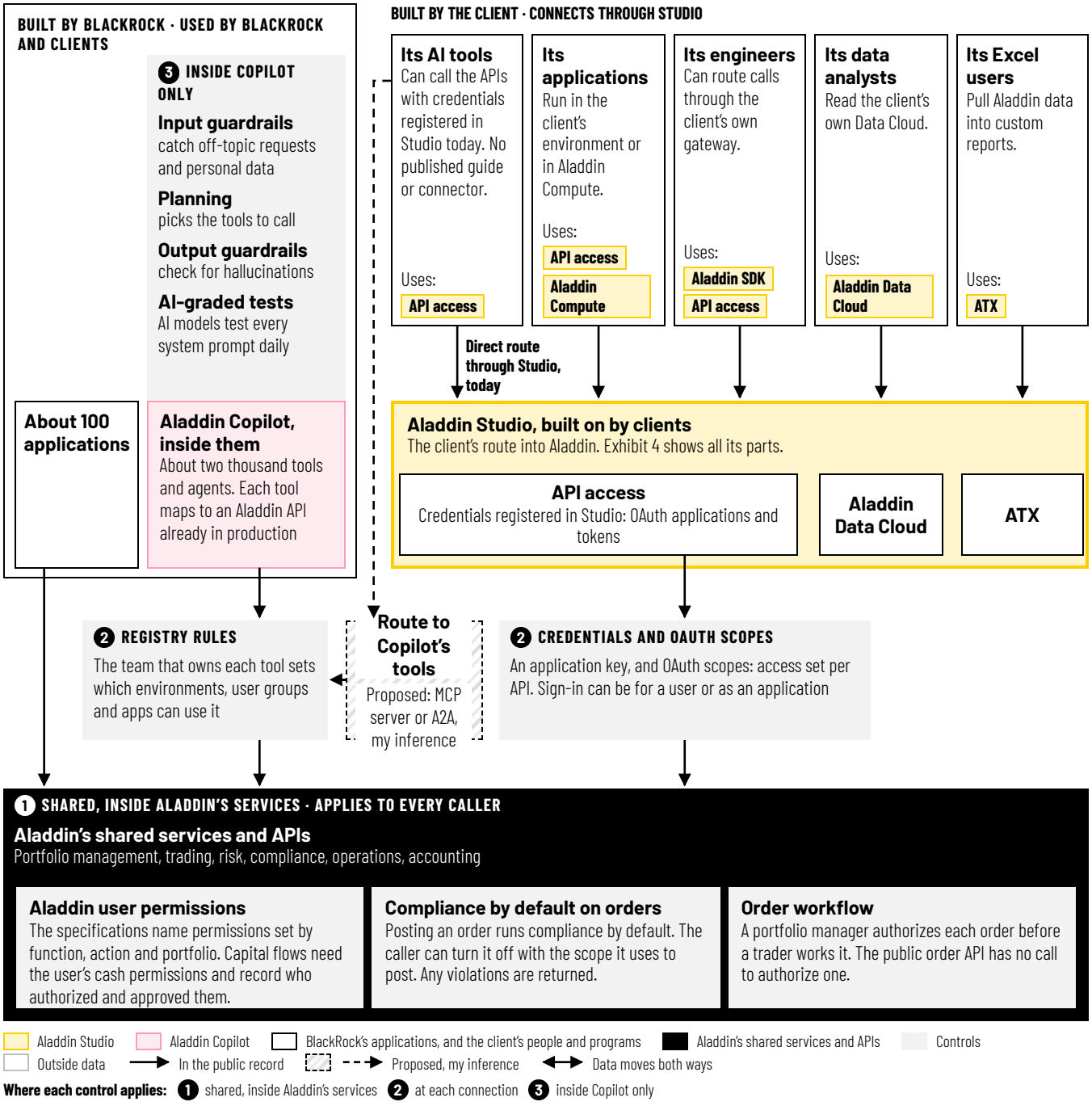
Three things would have to be true for this direction to hold. Clients would have to want AI tools that act, not only ones that answer questions. Their risk teams and regulators would have to accept it, which likely means a person approves each action. And BlackRock would have to open its controls to tools it did not build.

Exhibit 14. BlackRock's apps, Copilot and client code share Aladdin's APIs and controls. How a client's AI tools would connect is not public.



■ Solid boxes are in the public record. The hatched box is not. The nearest public sign is a September 2026 Forward Deploy Engineer posting for Aladdin Data that names MCP and AI agents[73]. The left-hand controls apply to calls through the APIs. Data Cloud, storage and the Data eXchange have their own access controls. The right-hand controls show how BlackRock governs AI on these APIs today, which is the model a client's own AI tools could follow. Sources: Rosales and Valdez[9], the Studio page[14], the Data Cloud launch release[34], the Copilot page[23], two SDK pull requests[56, 57], the SDK README[29], Aladdin's public API specifications[6], the Data Cloud page[20], Woodhead[1] and Fivetran[28].

Exhibit 14. Where each control applies: in Aladdin's shared services, at each connection, or inside Copilot only



For illustrative purposes only. Solid boxes and arrows are in the public record. The hatched box and the dashed arrow are my inference. That route would need BlackRock to apply Copilot's registry rules where a client's AI tools connect. Copilot's team has said it is evaluating A2A and LangChain's agent protocol. Whether Copilot's tools use the APIs clients can call is not public. The exhibit shows the controls on API calls only. Credentials registered in Studio, the gateway option and Aladdin Compute are as described in the public SDK, which may not match the current product. ■ Sources: Rosales and Valdez, for Copilot, its tools, registry rules, guardrails, planning, tests and A2A[9]; the Copilot page, for its use by clients[23]; the Studio page[14]; Aladdin's public API specifications, for the application key, scopes, sign-in, permissions, capital flows, compliance and the order workflow[6]; the public SDK's README and source, for credentials registered in Studio and Aladdin Compute[29]; two SDK pull requests, for the client's own gateway[56, 57]; the Data Cloud page and launch release, for each client's own Data Cloud[20, 34]; the ATX page[32].